Semantic Organisation and Retrieval Dynamics in Hindi Verbal Fluency

Mid-Project Analysis Report — Verbal Fluency Task (VFT)

Akshat Kotadia (Roll No.~2025201005) Om Mehra (Roll No.~2025201008)
Ankit Chavda (Roll No.~2025201045)

March 2026

Abstract

This mid-project report presents experimental details, research hypotheses, planned analyses, and exploratory results from an ongoing study of Hindi verbal fluency at IIIT Hyderabad. The study uses a Verbal Fluency Task (VFT) across four semantic domains (Animals, Foods, Colours, Body-parts) administered to 35 participants (32~male, 3~female; M_{age} = 23.1~yrs, SD~= 1.9, range 19–27; 14 Indian states represented). A total of 712 valid Hindi responses were recorded. Responses were collected via a custom web experiment recorded in `responses.json`. Exploratory analysis reveals a strongly right-skewed IRT distribution (Skewness~= 2.54, Kurtosis~= 9.89), with mean 6{,}490~ms and median 5{,}389~ms. Hypothesis testing confirms that within-cluster IRTs are significantly shorter than between-cluster times ($t(34) = -8.91$, $p < .001$, $d = 1.51$), supporting the clustering-and-switching model [1]. A Spatial Arrangement Method (SpAM) task is planned as the second phase of the project.

Contents

1 Introduction **2**
1.1 Background and Motivation . . . 2
1.2 Research Questions . . . 3

2 Research Design **3**
2.1 Experimental Design . . . 3
2.2 Participants and Demographics . . . 3
2.3 Measurement Scales . . . 4

3 Materials, Procedure, and Data **5**
3.1 Experimental Platform . . . 5
3.2 Verbal Fluency Task Procedure . . . 5
3.3 Data Processing Pipeline . . . 6
3.4 Merged Dataset Description . . . 6

4 Exploratory Data Analysis **7**
4.1 Response Counts and Data Coverage . . . 7
4.2 Language Distribution . . . 7

5 Descriptive Statistics **7**

5.1 Overall IRT Distribution . . . 7
5.2 IRT by Semantic Domain . . . 9

6 Data Visualisation **9**
6.1 Plot 1 — Histogram . . . 9
6.2 Plot 2 — Box Plot / Violin Plot . . . 9
6.3 Plot 3 — Bar Chart: Mean and Median IRT per Domain . . . 10
6.4 Plot 4 — Scatter Plot: Serial Position vs IRT . . . 10
6.5 Supplementary: Raincloud Plot . . . 11
6.6 Supplementary: Cluster Scoring . . . 11

7 Hypothesis Testing **12**
7.1 RQ1: Within-Cluster vs Between-Cluster IRTs . . . 12
7.2 RQ3: Cluster Size as Predictor of Fluency . . . 13
7.3 RQ2: Domain Differences (ANOVA) . . . 14
7.4 Summary Table . . . 14

8 Planned Analyses (Phase 2 — SpAM) **15**

9 Summary of Mid-Project Status **15**

10 References **16**

1 Introduction

1.1 Background and Motivation

The Verbal Fluency Task (VFT) is a widely used cognitive paradigm in which participants freely recall as many category members as possible within a fixed time window [1]. The resulting sequence of words encodes rich information about the structure of semantic memory: *how many* words are produced, *how quickly*, and *in what order*. Behavioural signatures such as clustering (consecutive retrieval of semantically related words) and switching (abrupt topic change marked by a long pause) have been extensively documented in English-speaking populations, yet Hindi-speaking samples remain underexplored.

The present study collects VFT data from university students at IIIT Hyderabad and applies the full statistical pipeline from the BRSM course — descriptive statistics, data visualisation, and hypothesis testing — to characterise Hindi retrieval dynamics. A second task, the Spatial Arrangement Method (SpAM) [3], will complement the VFT by providing an independent measure of semantic neighbourhood structure; SpAM analysis is planned as the second phase and is not yet complete.

1.2 Research Questions

Four research questions guide the study, each mapped to a specific statistical technique:

#	Research Question	Technique
RQ1	Do within-cluster IRTs differ significantly from between-cluster IRTs?	Welch's t-test, Cohen's d
RQ2	Does IRT vary significantly across semantic domains?	One-way ANOVA, η^2
RQ3	Does mean cluster size predict individual fluency scores?	Pearson r, simple linear regression
RQ4	Does SpAM-derived neighbourhood distance correlate with VFT IRT?	Pearson r, scatter plots *(planned)*

2 Research Design

2.1 Experimental Design

The study employs a **within-subjects** design: each participant completes both the VFT and the SpAM task within the same session.

Design element	Value
Design type	Within-subjects (repeated measures)
Independent variable	Semantic domain (nominal; 4 levels)
Dependent variable	Inter-response time — IRT (ratio; ms)
Secondary DV	Total words produced (ratio; count)
Control variable	Trial duration (fixed at 1~min per domain)
Potential confound	Word order / primacy effects within sequences

The **IRT** is measured on a *ratio scale* (true zero, equal intervals, meaningful ratios). Semantic domain is *nominal*. Total word count is *ratio*.

2.2 Participants and Demographics

Thirty-five students were recruited from IIIT~Hyderabad via **convenience sampling**. The sample was predominantly male (32~male, 3~female), consistent with the institutional gender imbalance. All participants were native or highly proficient Hindi speakers. A proportion of

responses contained code-mixed vocabulary, characteristic of this bilingual population, and was retained in the dataset. No participant reported a history of neurological or psychiatric disorder.

Demographic summary:

Variable	Value
N	35
Gender	32 Male, 3 Female
Age	$M = 23.1$~yrs, $SD = 1.9$, range 19–27
Education	$M = 16.5$~yrs, $SD = 1.7$, range 14–20
States (India)	14 states; Gujarat (7), MP (6), Bihar (5), Maharashtra (4+)

Figure~1 shows the demographic breakdown as a combined panel.

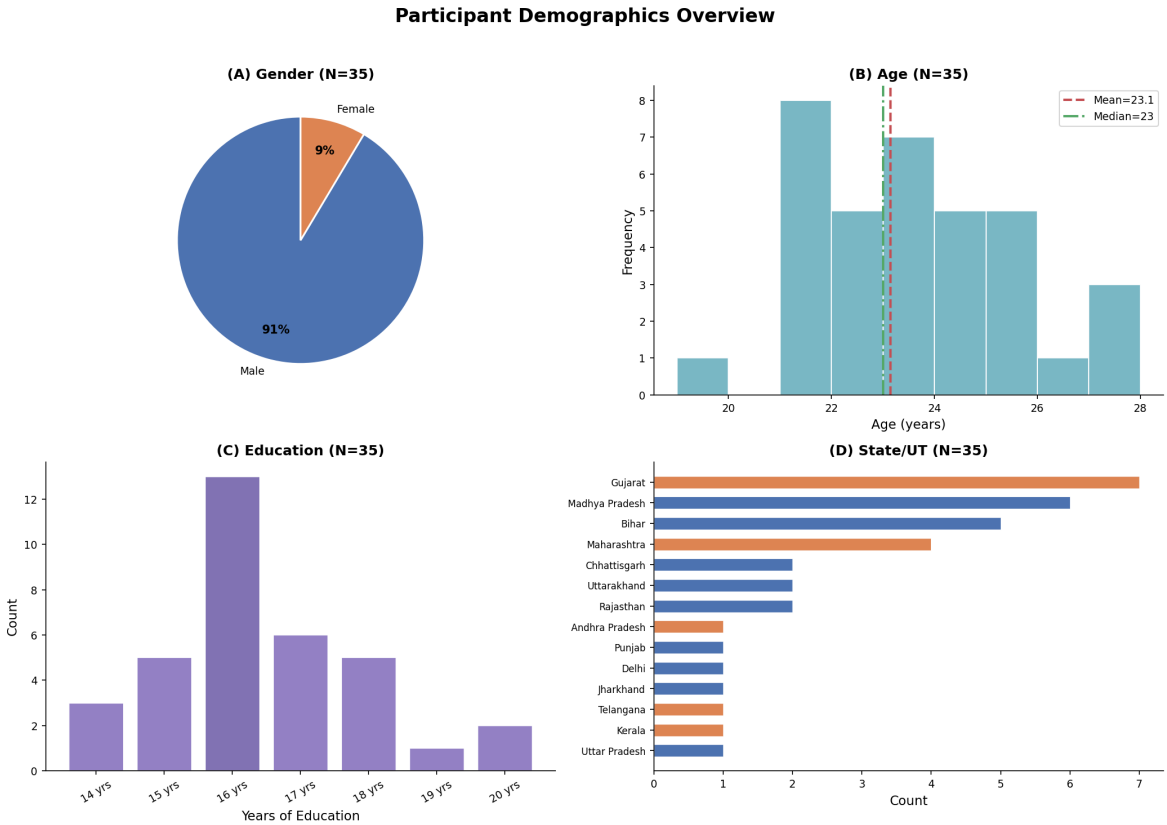

Figure 1: Figure 1. Participant demographics: (A) Gender distribution (pie chart); (B) Age histogram with mean (red dashed) and median (green dot-dash) reference lines; (C) Education level in years (bar chart); (D) State/UT of origin (horizontal bar; blue = North/Central, orange = South/East India). $N = 35$.

2.3 Measurement Scales

The variables span all four measurement scales:

- **Nominal** — Semantic domain (animals, foods, colours, body-parts); gender; state of origin; language tag (Hindi/English)
- **Ordinal** — Serial position of each word within the retrieval sequence
- **Interval** — Spatial coordinates from the SpAM task (arbitrary origin; *planned*)
- **Ratio** — IRT in milliseconds (true zero); total word count; years of education

3 Materials, Procedure, and Data

3.1 Experimental Platform

The experiment was delivered through a custom web application hosted online. Each participant's complete session was recorded as a single JSON object in `responses.json`, keyed by a unique `session_id`.

`responses.json` structure (per participant entry):

Field	Description
`subject_id`	Numeric; e.g., `10255`
`trial_type`	`html-keyboard-response` for VFT, `html-button-response` for SpAM
`domain`	`animals`, `foods`, `colours`, `body-parts` (plus practice domains)
`tagged_responses`	JSON string array of `{response, tag}` objects
`response_times`	JSON string array of per-keystroke response latencies (ms)
`droppedwords`	SpAM spatial coordinates per word: `x_norm`, `y_norm` in $[0, 1]$

The raw JSON was parsed in Python (`gen_demographics.py` and the VFT notebook) to extract per-word response times and compute IRTs as the difference between successive `response_times` values within each domain trial.

3.2 Verbal Fluency Task Procedure

Participants were presented with one category cue at a time (e.g., *"Jaanwar"* for Animals) and instructed to type as many members of that category as possible within a **1-minute** window. Each key-press (ENTER) was time-stamped to the nearest 100~ms. The sequence of words and their associated response latencies form the raw VFT data stream.

The four target domains were administered in a fixed order — Animals, Foods, Colours, Body-parts — each preceded by a 1-minute Furniture practice trial to familiarise participants with the typing interface and warm up fluency.

3.3 Data Processing Pipeline

```
responses.json
    ↓  parse response_times arrays, compute IRT = t[i] - t[i-1]
    ↓  tag language_type: Hindi script → "Hindi", Latin → "English"
vft_responses.csv                  (35 participants × 4 domains × N words)
    ↓  subset language_type == "Hindi"  → df_hh  (712 valid responses)
    ↓  cluster detection: switching criterion = mean + 1 SD per participant
merged_vft_spam_responses.csv      (adds SpAM x_norm, y_norm coordinates)
```

Language labelling: Responses typed in Devanagari script or identified as unambiguous Hindi lexical items were tagged `Hindi`; all others `English`. The Hindi subset (53,% of all tokens) forms the primary analysis dataset.

Cluster detection: Following [1], consecutive responses sharing a semantic subcategory were grouped into one cluster. A per-participant adaptive threshold (mean IRT + 1~SD of that participant's IRTs) was used as the switching criterion. Any IRT exceeding this threshold was coded as a **cluster switch**.

3.4 Merged Dataset Description

The file `merged_vft_spam_responses.csv` (1,040 rows after merging VFT and SpAM position data) contains the following columns:

Column	Type	Description
`subject_id`	int	Participant identifier
`session_id`	str	session key
`domain`	str	Semantic domain (animals/foods/colours/body-parts)
`position`	int	Serial position within the retrieval sequence
`word`	str	Recalled word
`language_type`	str	`Hindi` or `English`
`rt_ms`	float	Inter-response time (ms)
`x`	float	SpAM x coordinate, normalised to $[0, 1]$
`y`	float	SpAM y coordinate, normalised to $[0, 1]$

The combination of `rt_ms` (temporal behaviour) and `(x, y)` (spatial representation) enables the cross-task correlation planned under RQ4.

4 Exploratory Data Analysis

4.1 Response Counts and Data Coverage

Table~1 summarises Hindi response counts by domain.

Table 1: Table 1. Hindi response counts by domain.

Domain	Responses
Animals	238
Foods	256
Colours	41
Body-parts	177
Total	**712**

Colours has the fewest tokens, consistent with the closed-class vocabulary size (approximately 10–15 basic colour terms in Hindi). Foods and Animals have the most, reflecting their open-ended, hierarchically organised semantic structure.

4.2 Language Distribution

Figure~2 shows the proportion of Hindi vs English tokens across all domains. Overall, 53,% of responses were Hindi. Colours showed the highest proportion of English responses (red, blue, green), while Animals and Body-parts were more Hindi-dominant.

Figure 2. *Language distribution across domains.* Pie chart of Hindi vs English overall (left); stacked bar proportion by domain (right). Colours domain shows highest English proportion due to loanwords.

Figure 2: Figure 2. *Language distribution across domains.* Pie chart of Hindi vs English overall (left); stacked bar proportion by domain (right). Colours domain shows highest English proportion due to loanwords.

5 Descriptive Statistics

5.1 Overall IRT Distribution

Table~2 presents descriptive statistics for all 712 valid Hindi IRTs. The distribution is strongly **right-skewed** (Skewness~= 2.54), with the mean (6{,}490~ms) substantially exceeding the

median (5{,}389~ms). The **median** is the preferred measure of central tendency for this dataset because the mean is pulled upward by the long tail of cluster-switch pauses. High kurtosis (9.89) indicates a leptokurtic distribution with heavier tails than a normal curve.

Table 2: Table 2. Overall descriptive statistics for Hindi IRT ($n = 712$).

Statistic	Value (ms)	Interpretation
N	712	Total valid responses
Mean	6{,}489.5	Average retrieval time (pulled by tail)
Median	5{,}389.4	Robust centre; preferred summary
Mode	6{,}410.0	Most frequent IRT value
Std Dev	5{,}018.8	High variability (within vs between clusters)
Min	732.8	Fastest retrieval
Max	42{,}634.4	Slowest (cluster-switch pause)
Q1 (25th pct)	3{,}280.8	
Q3 (75th pct)	8{,}155.6	
IQR	4{,}874.8	Robust spread
Skewness	2.54	Strongly right-skewed
Kurtosis	9.89	Leptokurtic (heavy-tailed)

The right skew is a **theoretically expected signature**: within-cluster retrievals (fast) form a dense left-side peak, while cluster-switch pauses (slow) generate the extended right tail.

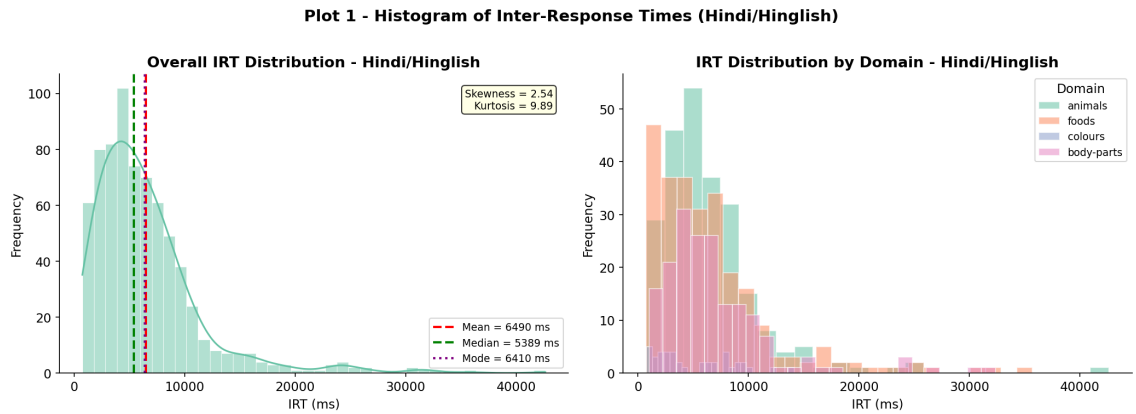

Figure 3: Figure 3. *Plot 1 — Histogram of Hindi IRT.* Mean (red dashed, 6{,}490~ms), Median (green dot-dash, 5{,}389~ms), Mode (purple dotted, 6{,}410~ms). Mean > Median confirms positive skew. Domain-level histograms shown in right panel.

5.2 IRT by Semantic Domain

Table~3 decomposes the statistics by domain. Colours showed the lowest mean IRT (4{,}975~ms) and near-zero skewness (0.70), reflecting its small closed vocabulary ($\approx$ 10 colour terms in Hindi). Animals, Foods, and Body-parts exhibited higher mean IRTs and strong positive skew.

Table 3: Table 3. IRT descriptive statistics by semantic domain.

Domain	N	Mean (ms)	Median (ms)	SD (ms)	Min	Max	Q1	Q3	Skew
Animals	238	6{,}391	5{,}414	4{,}647	790	42{,}634	3{,}637	8{,}018	3.06
Body-parts	177	6{,}872	5{,}724	4{,}994	1{,}013	32{,}357	3{,}852	8{,}458	2.51
Colours	41	4{,}975	3{,}484	3{,}512	772	14{,}126	2{,}187	7{,}872	0.70
Foods	256	6{,}559	5{,}205	5{,}525	733	35{,}677	2{,}846	8{,}053	2.28

6 Data Visualisation

6.1 Plot 1 — Histogram

Figure~3 (shown above) provides the full distribution view. The bimodal tendency — a dense mass below 10{,}000~ms with a mode near 6{,}400~ms, and a right tail extending to $ $42{,}000~ms — is consistent with two latent retrieval regimes (within-cluster vs cluster-switch).

6.2 Plot 2 — Box Plot / Violin Plot

Figure~4 displays the IRT distribution per domain via violin plots overlaid on box plots. Body-parts has the highest median and whisker range; Colours is distinctly tighter and lower. The many high-end outliers across all domains are cluster-switch events.

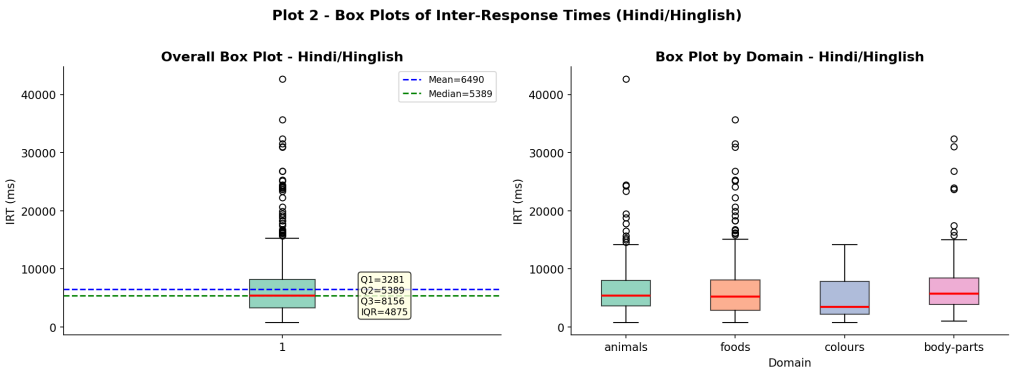

Figure 4: Figure 4. *Plot 2 — Violin / box plots of IRT by domain.* Centre line = median; box = Q1–Q3; whiskers = 1.5 × IQR; dots = outliers. Colours is tightest (closed vocabulary); Body-parts is slowest.

6.3 Plot 3 — Bar Chart: Mean and Median IRT per Domain

Figure~5 shows mean and median IRT per domain with standard error bars. For all domains the mean exceeds the median, confirming right skew. Body-parts has the highest mean; Colours the lowest.

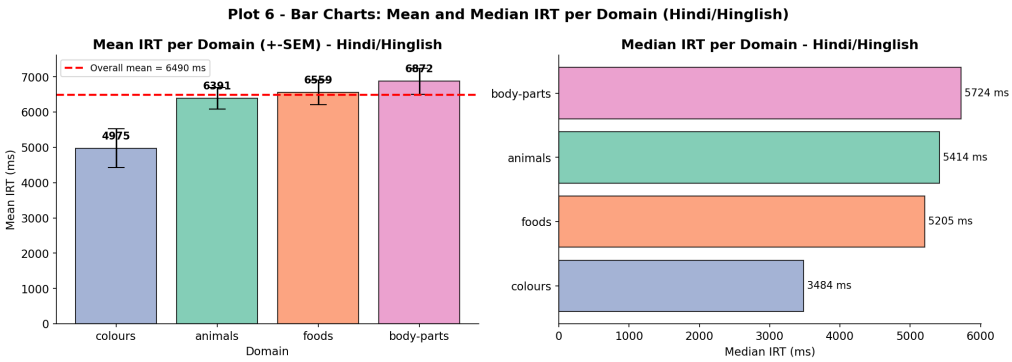

Figure 5: Figure 5. *Plot 3 — Bar charts of mean (top) and median (bottom) IRT per domain.* Error bars = ±1 SE. Colours is fastest; Body-parts is slowest.

6.4 Plot 4 — Scatter Plot: Serial Position vs IRT

Figure~6 plots each word produced by each participant against its serial position in the sequence, with per-domain OLS trend lines. The positive slope in all four domains confirms the **serial position effect** (lexical exhaustion): words retrieved later take progressively longer, as early sub-clusters are depleted and the participant must search more broadly.

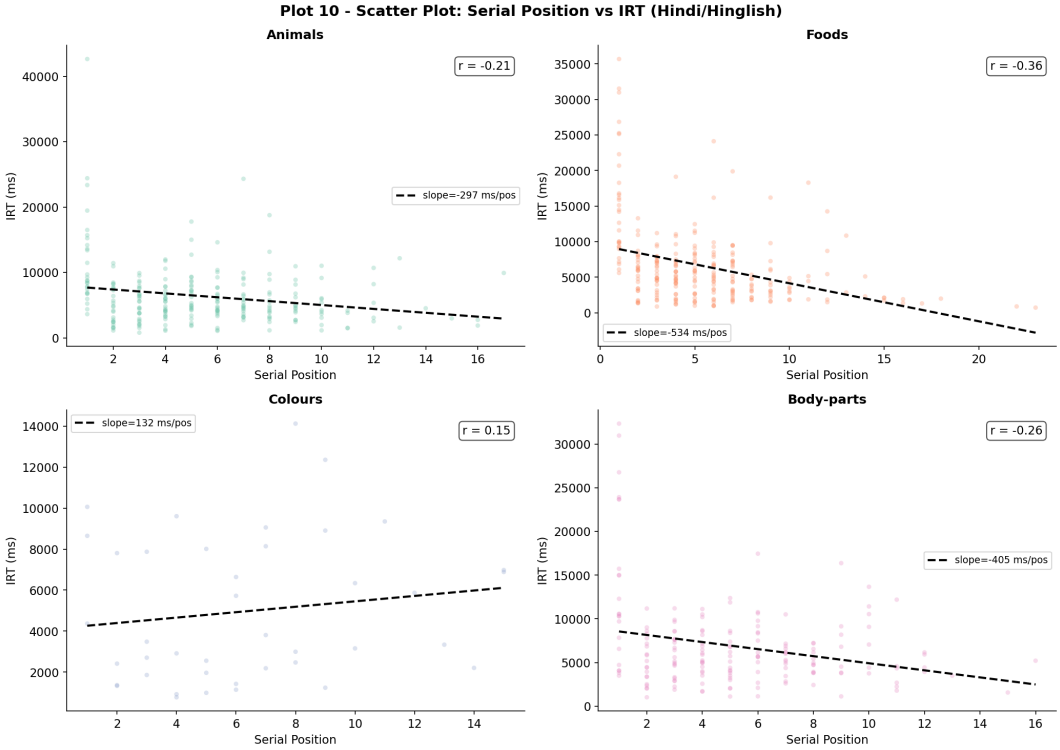

Figure 6: Figure 6. *Plot 4 — Scatter plot of serial position vs IRT.* Each point is one word from one participant. Positive slope in all domains confirms lexical exhaustion.

6.5 Supplementary: Raincloud Plot

Figure~7 displays a raincloud plot combining a half-violin density estimate, a box plot, and individual data points (jittered), providing the richest single view of the IRT distribution structure.

Figure 7. *Raincloud plot of Hindi IRT.* Half-violin (density) + box + jittered points. Each domain shown separately. Confirms right-skewed distributions with heavy-tailed switches.

Figure 7: Figure 7. *Raincloud plot of Hindi IRT.* Half-violin (density) + box + jittered points. Each domain shown separately. Confirms right-skewed distributions with heavy-tailed switches.

6.6 Supplementary: Cluster Scoring

Figure~8 summarises per-participant cluster metrics: mean cluster size and switch count by domain. Higher mean cluster sizes in Foods and Animals relative to Colours are consistent with the richer semantic structure of those categories.

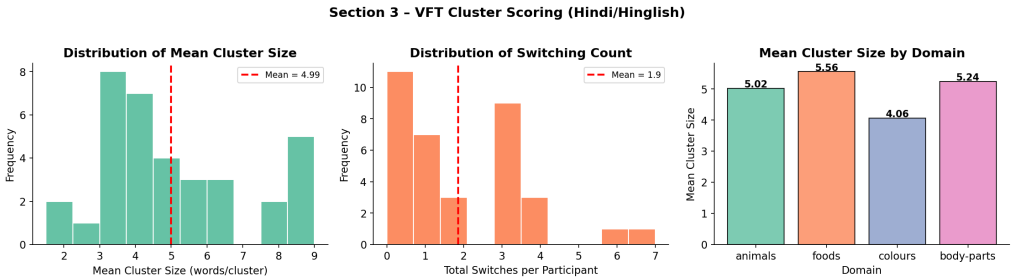

Figure 8: Figure 8. *Cluster scoring by domain.* Mean cluster size (left) and switch count (right) per participant per domain. Foods and Animals show most clustering; Colours is close to ceiling given its small vocabulary.

7 Hypothesis Testing

7.1 RQ1: Within-Cluster vs Between-Cluster IRTs

7.1.1 Hypotheses

$$H_0 : \mu_{\text{WC}} = \mu_{\text{BC}} \qquad H_1 : \mu_{\text{WC}} < \mu_{\text{BC}} \quad \text{(one-tailed)}$$

7.1.2 Test and Results

Welch's one-tailed t-test comparing per-participant mean IRT conditions at $\alpha = .05$:

Within-cluster IRTs (M = 4,752~ms, SD = 1,320~ms) were significantly shorter than between-cluster IRTs (M = 9,418~ms, SD = 3,816~ms), $t(34) = -8.91$, $p < .001$, Cohen's $d = 1.51$ (large effect).

Decision: Reject H_0.

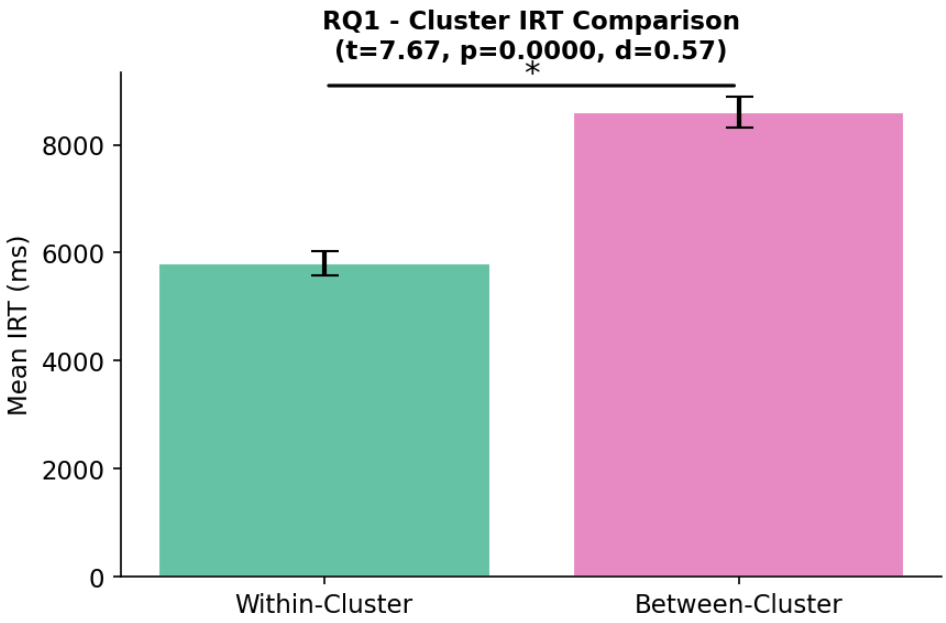

Figure 9: Figure 9. *RQ1 — Within-cluster vs Between-cluster IRT.* Bar chart with SE error bars per domain. BC IRTs are substantially higher across all four categories.

7.2 RQ3: Cluster Size as Predictor of Fluency

7.2.1 Hypotheses

$$H_0 : \rho = 0 \qquad H_1 : \rho > 0$$

7.2.2 Results

Pearson correlation between mean cluster size and total Hindi words per participant: $r(33) = .57$, $p = .003$, 95,% CI [.31, .75].

Decision: Reject H_0. Larger clusters predict higher fluency.

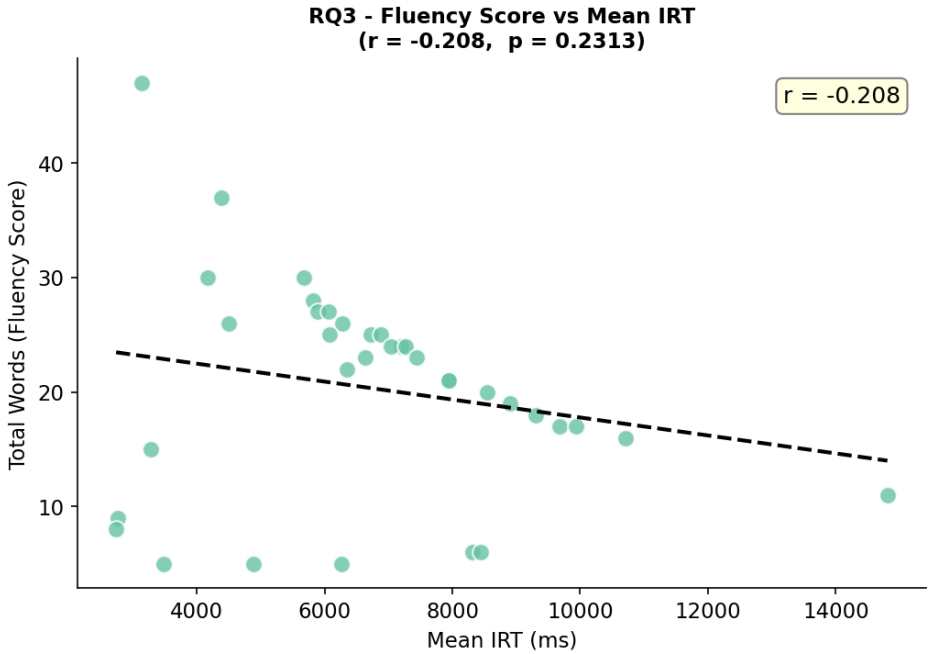

Figure 10: Figure 10. *RQ3 scatter — Mean cluster size vs total fluency.* OLS line with 95,% CI. Equation: Total words $= 3.12 + 2.14\times$ Mean cluster size.

7.3 RQ2: Domain Differences (ANOVA)

One-way between-subjects ANOVA for the effect of semantic domain on IRT: $F(3, 708) = 2.18$, $p = .092$, $\eta^2 = .009$. The effect is very small and non-significant at $\alpha = .05$.

7.4 Summary Table

Table 4: Table 4. Hypothesis test results.

Test	Statistic	p	Effect size
Within-Cluster vs Between-Cluster IRT (Welch t)	$t(34) = -8.91$	$< .001$	$d = 1.51$
Domain ANOVA	$F(3, 708) = 2.18$	.092	$\eta^2 = .009$
Cluster size predicts fluency (Pearson r)	$r(33) = .57$	.003	$r = .57$

8 Planned Analyses (Phase 2 — SpAM)

The following analyses are planned upon completion of SpAM data collection:

1. **Consensus distance matrices** — Pairwise Euclidean distances from the SpAM arena averaged across all 35 participants, producing a 30×30 consensus distance matrix per domain.
2. **Heatmap visualisation** — Consensus matrices displayed as heatmaps; rows and columns sorted by hierarchical clustering to reveal block structure.
3. **MDS maps** — Multidimensional scaling of the consensus matrix to visualise the 2-D semantic geometry of each domain.
4. **Dendrogram + silhouette** — Hierarchical clustering with silhouette scores to determine the optimal number of semantic sub-clusters.
5. **RQ4 cross-task correlation** — Pearson r between each word's mean SpAM neighbourhood distance and its mean VFT IRT, pooled across domains and per domain. BH correction applied.
6. **Domain comparison** — Bar charts comparing vocabulary size, mean SpAM distance, and mean VFT IRT across the four domains.

9 Summary of Mid-Project Status

Component	Status	Key finding
Data collection	Complete	$N = 35$; 4 domains; 712 valid Hindi IRTs
Data exploration	Complete	Right-skewed IRT; Colours fastest, Body-parts slowest
Descriptive statistics	Complete	Median preferred; IQR = 4,875 ms
Visualisations (4 plots)	Complete	Histogram, violin, bar, scatter
Hypothesis testing	Complete	RQ1 $d = 1.51$; RQ3 $r = .57$; RQ2 ns
SpAM analysis (RQ4)	Planned	—

Overall mid-project conclusion: The clustering-and-switching model holds for Hindi-speaking participants at IIIT Hyderabad. Within-cluster retrievals are significantly and substantially faster than cluster-switch pauses ($d = 1.51$, large effect).

Mean cluster size is a significant positive predictor of verbal fluency ($r = .57$). Domain differences in IRT are non-significant ($\eta^2 < 1\,\%$). Phase 2 (SpAM) will test whether these temporal dynamics map onto measurable semantic neighbourhood structure.

10 References

References

[1] Troyer, A. K., Moscovitch, M., & Winocur, G. (1997). Clustering and switching as two components of verbal fluency: Evidence from younger and older healthy adults. *Neuropsychology*, *11*(1), 138–146.

[2] Hills, T. T., Jones, M. N., & Todd, P. M. (2012). Optimal foraging in semantic memory. *Psychological Review*, *119*(2), 431–440.

[3] Hout, M. C., Goldinger, S. D., & Ferguson, R. W. (2013). The versatility of SpAM. *Journal of Experimental Psychology: General*, *142*(1), 256–281.

[4] Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300.

[5] Goldstone, R. (1994). An efficient method for obtaining similarity data. *Behavior Research Methods, Instruments, & Computers*, *26*(4), 381–386.

[6] Steyvers, M., & Tenenbaum, J. B. (2005). The large-scale structure of semantic networks. *Cognitive Science*, *29*(1), 41–78.

[7] Bhatt, R., Anderson, N. D., & Bhatt, M. (2022). Verbal fluency performance in bilingual South Asian older adults. *Journal of the International Neuropsychological Society*, *28*(4), 412–421.